

COURSE NAME : PRESS TOOL TECHNOLOGY
COURSE CODE : 5111
SEMESTER :5
COURSE CATEGORY : A
PERIODS / WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Sheet metal processing using Press Tool Punches, Dies and Stoppers	15
2	Strippers, gauges, Pilots, side cutters, ejectors and Knockouts, Die sets and Progressive tool, shank, Presses and Strip feeding	15
3	Compound Tool Bending Tool	15
4	Forming and Drawing Tool Fine Blanking	15
	Total	60

COURSE OUTCOMES

On completion of the study the students will be able to :

1. Understand the sheet metal processing using different types of press tools
2. Understand Punches, Dies and Stoppers.
3. Understand the parts of press tools.
4. Know Progressive Tools and Die sets.
5. Understand working of presses.
6. Understand working of compound tools.
7. Know the bending tool.
8. Understand Forming and Drawing operations.
9. Know the Fine blanking process.

SPECIFIC OUTCOMES

MODULE I

1.1.0 Understand Sheet Metal Processes using different types of Press Tools

- 1.1.1 List out the Press Tool operations.
- 1.1.2 Draw different strip layout.
- 1.1.3 Estimate percentage utilization and economy factor of strip.

1.2.0 Comprehend Punches, Dies and stoppers

- 1.2.1 List the type of punches.
- 1.2.2 State the type of die block.
- 1.2.3 Identify different type of stoppers.

MODULE II

2.1.0 Analyze the parts of Press Tool

- 2.1.1 Describe the functions of strippers.
- 2.1.2 State the dimensional aspect of gauges.
- 2.1.3 List the use of pilots and side cutters.
- 2.1.4 Compare ejectors and knockouts.

2.2.0 Know the progressive tool and Die sets

- 2.2.1 Calculate the shank location.
- 2.2.2 Identify the type of die sets.
- 2.2.3 List out the function of progressive tool.

2.3.0 Understand the working of Presses

- 2.3.1 List the parts of press.
- 2.3.2 State the classification of presses.
- 2.3.3 Illustrate the strip feed arrangements on press.

MODULE III

3.1.0 Understand the working of compound tool

- 3.1.1 State the elements of compound tool.
- 3.1.2 Compare progressive and compound tool.
- 3.1.3 Describe shaving operation.

3.2.0 Know the bending tool

- 3.2.1 Explain Bending elements of V and U Bending dies.
- 3.2.2 List the bending tools.
- 3.2.3 State the effect of grain direction in bending.

MODULE IV

4.1.0 Understand the Forming and Drawing Operations

- 4.1.1 Categorize different forming operations.
- 4.1.2 Identify faults in drawing and remedies.

4.2.0 Know Fine Blanking Process

- 4.2.1 Describe the application and working of fine blanking.
- 4.2.2 State punch and die radius.
- 4.2.3 Explain fine blanking tool.
- 4.2.4 Identify materials suitable for fine blanking.

CONTENT DETAILS

MODULE I

Press Tool operations

Introduction to piercing tool, blanking tool, progressive tool, compound tool, cut off tool,

parting tool, notching tool, trimming tool, shaving tool, performing tool, lancing tool, dinking tool, Side cam tool, broaching tool, extrusion tool, planishing tool, embossing tool, coining tool, bending tool, forming tool, drawing tool, curling tool, bulging tool, swaging tool, combination tool, assembly tool.

Strip layout – narrow run-wide run - single row – double row – gang dies – grain direction —

layout for blanking tool, progressive tool.

Economy factor, calculation of percentage utilization of strip.

Punches – cutting – non cutting – Hybrid punches – types of punches – plain – pedestal – offset

Pedestal– headless – step head – beveled head– perforators - material of punch.

Buckling of punches, buckling of theorem – problems – type of loads on punch, determining the size of punch.

Die blocks – types – solid – split, sectional– selection of die block, Material

for die block, Stoppers – functions – position – stop categories – primary – secondary final, Construction – pin stop – headed pin stop – disappearing pin stop – finger stop – pusher stop

trigger stop.

MODULE II

Strippers – function, types-fixed and traveling. Constructional details – spring stripper,

clamping spring, , spring around stripper bolt. Stripper bolt suspension, Stripping force for blanking & piercing

Gauges (Strip guide), Functions – dimensional aspect of gauge, Types – back, front, bulge clearance extended back gauge – strip support – pushers, nesting gauges, fool proofing – strip lifters Pilots, Purpose, indirect and direct pilot , size, length of pilot, Pilot opening in die and in die shoe,

Types – functions of different types of pilots, Methods of piloting

Side cutters - Function - advantages

Ejectors and Shedders, Function - types

Knockouts, Function, - direct & indirect knockouts

Fasteners and dowels, Dowel – function – types, Fasteners – screws – function – types of screws like socket head, counter sunk, cheese head, set screw fasteners, Rivets, cotter pins – positioning of screw progressive tool and die sets. Shank , Types , location- arithmetical and graphic method.

Die sets, Types – precision and commercial / standard & non standard.

Components – top plate – bottom plate, guide Pillar. Define Shut height, functions of

Progressive tool. Construction and related design. Presses

Parts, capacity, bed, slide, bolster plate, pitman, clutch, stroke, shut height, die space, capacity, number of slides, actuation of slide.

Press frame types , clutch , brake, stroke adjustments.

Classification of press – source of power – mechanical, hydraulic& pneumatic.

Strip feed arrangements on press.

MODULE III

Compound tool, Function of elements – related design, Progressive and compound tool – advantages disadvantages, Shaving – process – purpose, Clearance – punch and die size.

Bending tools – principle – plastic deformation, Bending elements, Blank length – calculation of strip required, Bending elements of V and U bending, Effect of grain direction in bending.

bending in proximity to pierced holes, Material for bending

MODULE IV

Different forming operations, Embossing, coining, curling, flanging, planishing, drawing, deep drawing, cylindrical cup drawing – Wrinkling and puckering.

Blank development for cylindrical shell – necessity – Algebraic method – graphical method – layout methods – center of gravity method.

Fine Blanking, Application – working principle, V-Ring – function, dimension, Strip width and margin – calculation, Importance of Punch and die radius, Materials suitable for fine blanking work. Fine blanking tool working, clearance calculation – piece part configuration – radii on sharp corners – length and width of slots and webs – ring width hole distance and web width – tool module.

REFERENCE BOOKS

1. Tool design - Donaldson
2. Die design fundamentals - Paquin
3. Basic die making - Ostergarrd
4. Advanced die making - Ostergarrd
5. Fundamentals of tool design - ASTM E
6. Tool design - C.B. Cole
7. Tool engineers parameters - CITD
8. A Text Book of Production Engg. - P. C. Sharma (S.Chand)
9. Die Design Handbook – ASME
10. A Textbook of Machine Tool & Tool Design - P. C. Sharma